

Scenario: The Bank Loan & Interest Rate Approver

A local bank needs an automated system to evaluate loan applications and calculate the approved interest rate. The decision relies on preliminary eligibility checks, credit scores, income ratios, and collateral.

Nested conditions are necessary here because the bank only checks financial details (like income or collateral) **after** an applicant meets baseline eligibility rules (like age and employment status).

Student Assignment: Loan Engine Challenge

Write a program that determines if an applicant is approved for a loan and calculates their final annual interest rate based on their profile.

Inputs

- `age` (integer)
- `is_employed` (boolean)
- `credit_score` (integer)
- `annual_income` (float)
- `has_collateral` (boolean)

Business Logic & Rules

1. **Baseline Eligibility (Outer Condition)**
 - Applicant must be at least **21 years old** AND currently **employed**.
 - *If they fail this check:* Instantly reject the loan with an output message: "Rejected: Fails baseline criteria".
2. **Financial Evaluation (Inner/Nested Conditions)**
 - If baseline eligibility is met, evaluate their `credit_score`:
 - **Tier 1: High Credit (`credit_score >= 750`)**
 - Base interest rate: **5.0%**
 - If `annual_income >= 100,000`, give a loyalty discount: final rate is **4.5%**.
 - **Tier 2: Fair Credit (`600 <= credit_score < 750`)**
 - Base interest rate: **8.0%**
 - If `has_collateral` is `True`, reduce the rate to **7.0%**.
 - Otherwise, if `annual_income < 40,000`, increase the risk rate to **9.5%**.
 - **Tier 3: Low Credit (`credit_score < 600`)**
 - Reject the loan with output message: "Rejected: Credit score too low".

Expected Test Cases

Case	Age	Employed?	Credit Score	Income	Collateral?	Expected Output
1	19	True	780	\$120,000	False	Rejected: Fails baseline criteria
2	30	True	780	\$110,000	False	Approved at 4.5% interest
3	45	True	680	\$50,000	True	Approved at 7.0% interest

4	28	True	620	\$35,000	False	Approved at 9.5% interest
5	35	True	550	\$80,000	True	Rejected: Credit score too low
6				40000	True	